HealthRisk AI Dashboard

Technical Project Report

Submitted By

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Project Domain

Artificial Intelligence, Machine Learning and Healthcare Analytics

Contents

Abstract

HealthRisk AI Dashboard is a Machine Learning based healthcare analytics platform developed using Python and Streamlit.

The system integrates multiple healthcare and financial risk prediction modules into a unified dashboard.

The project consists of:

- Heart Disease Prediction
- Insurance Risk Assessment
- Hospital Risk Analysis
- Pharmaceutical Recommendation System
- HealthRisk Lab Simulation Environment

The objective is to assist healthcare decision-making through predictive analytics and interactive visualization.

Chapter 1

Introduction

1.1 Background

Healthcare organizations generate massive volumes of data including clinical records, insurance claims, pharmaceutical research data, and operational information.

Artificial Intelligence enables healthcare institutions to extract actionable insights from these datasets.

1.2 Problem Statement

Traditional healthcare analysis is often manual and time-consuming.

The project aims to develop an AI-powered platform capable of predicting healthcare risks and providing decision support.

1.3 Objectives

- Predict heart disease risk
- Assess insurance risk
- Evaluate hospital financial risk
- Generate pharmaceutical recommendations
- Develop an interactive healthcare dashboard

Chapter 2

Literature Review

2.1 Artificial Intelligence in Healthcare

Artificial Intelligence is increasingly used for disease prediction, healthcare management, and medical decision support.

2.2 Machine Learning Applications

Applications include:

- Disease prediction
- Insurance analytics
- Hospital risk management
- Pharmaceutical analytics

2.3 Dashboard Based Decision Support

Interactive dashboards improve data interpretation and support informed decision-making.

Chapter 3

System Architecture

3.1 Overall Architecture

User Interface → Preprocessing → Machine Learning Models → Prediction Engine → Visualization

3.2 Architecture Diagram

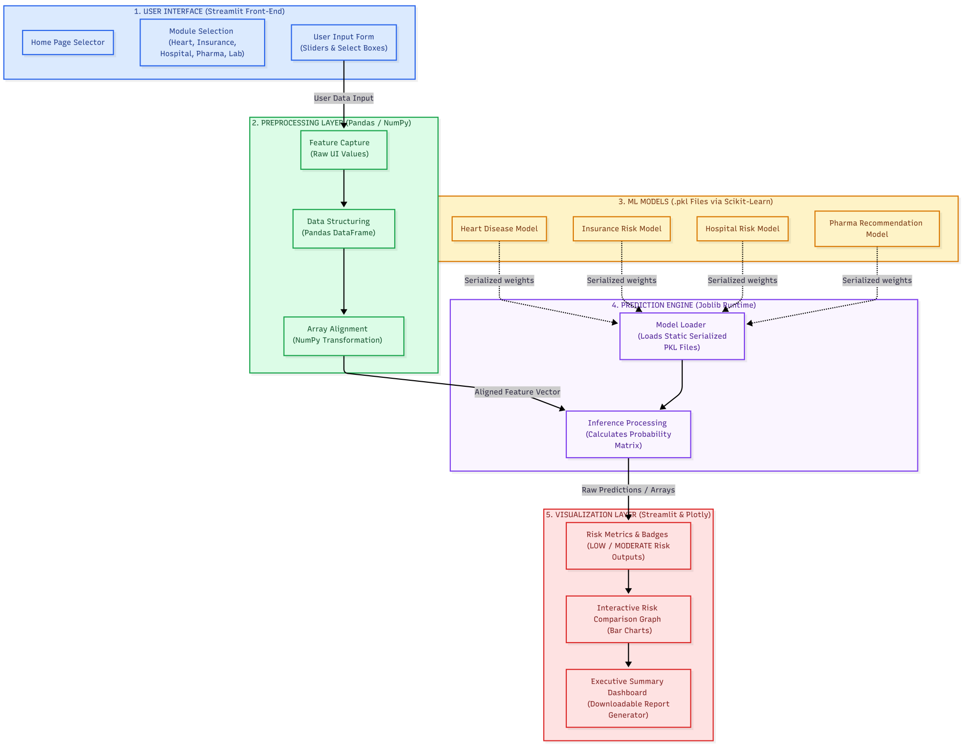

Figure 3.1: HealthRisk AI System Architecture

Chapter 4

Technologies Used

4.1 Programming Language

- Python

4.2 Libraries

- Pandas
- NumPy
- Scikit-Learn
- Streamlit
- Joblib
- Matplotlib
- Seaborn

Chapter 5

Dataset Description

5.1 Heart Disease Dataset

Features:

- Age
- Sex
- Chest Pain Type
- Cholesterol
- Blood Pressure
- ECG
- Heart Rate

Target: Heart Disease

5.2 Insurance Dataset

Features:

- Age
- BMI
- Smoker
- Region
- Children

Target: Risk Category

5.3 Hospital Dataset

Features:

- Revenue
- Debt Ratio
- Readmission Rate
- Patient Satisfaction

Target: Risk Level

Chapter 6

Data Preprocessing

6.1 Data Cleaning

- Missing Value Handling
- Duplicate Removal
- Outlier Detection

6.2 Feature Engineering

- Label Encoding
- Feature Scaling
- Train-Test Split

Chapter 7

Machine Learning Models

7.1 Heart Disease Prediction

Algorithm: Random Forest Classifier

7.2 Insurance Risk Prediction

Algorithm: Random Forest Classifier

7.3 Hospital Risk Prediction

Algorithm: Random Forest Classifier

7.4 Pharma Recommendation

Algorithm: Decision Tree Classifier

Chapter 8

Dashboard Development

8.1 Streamlit Dashboard

Modules:

1. Home Page
2. Heart Disease Prediction
3. Insurance Risk Prediction
4. Hospital Risk Assessment
5. Pharma Recommendation
6. HealthRisk Lab Simulator

8.2 Workflow

1. User enters data
2. Dashboard processes input
3. Model generates prediction
4. Result displayed

Chapter 9

Results and Testing

9.1 Model Evaluation

Example Table:

Model	Accuracy	Precision	Recall
Heart Disease	88%	91%	90%
Insurance Risk	89%	88%	87%
Hospital Risk	90%	89%	88%
Pharma Recommendation	90%	85%	84%

Table 9.1: Model Performance

9.2 Dashboard Screenshots

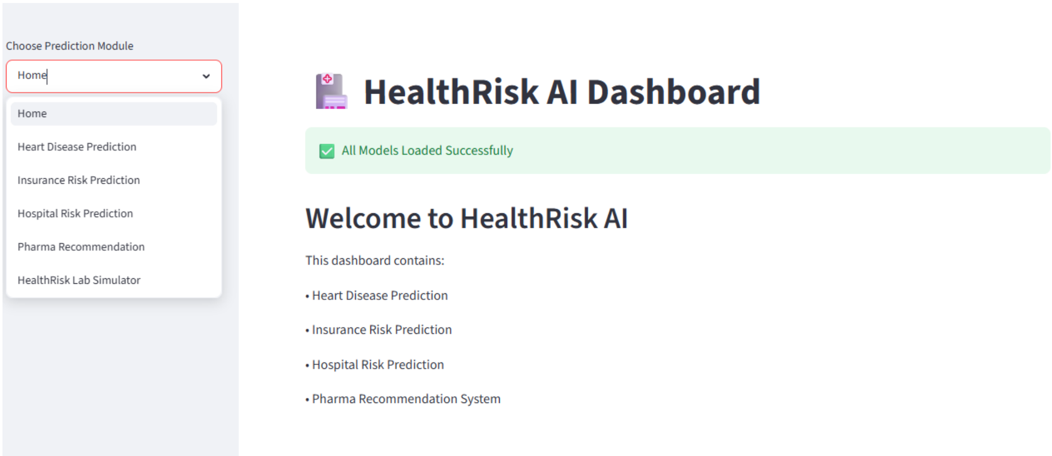

Figure 9.1: Dashboard Home Page

Chapter 10

HealthRisk Lab Simulation

The simulation environment demonstrates the effect of healthcare events on risk levels and financial outcomes.

Example events:

- Pandemic Outbreak
- Insurance Claim Surge
- Drug Trial Success
- Hospital Revenue Decline

Chapter 11

Advantages

- User Friendly
- Scalable
- Multiple Risk Models
- Interactive Dashboard
- Educational Value

Chapter 12

Limitations

- Limited Real-Time Data
- Small Dataset Dependency
- Requires Model Retraining

Chapter 13

Project Deployment and Demonstration

13.1 Deployed Application

The HealthRisk AI Dashboard has been deployed online using Streamlit Cloud, allowing users to access the application through a web browser without requiring local installation.

Application URL:

https://healthriskai-fbvjkytvktmxkwrxcmuy8w.streamlit.app/

The deployed application includes all major modules:

- Heart Disease Prediction
- Insurance Risk Assessment
- Hospital Risk Assessment
- Pharma Recommendation System
- HealthRisk Lab Simulator

13.2 Demonstration Video

A complete demonstration video has been prepared to showcase the functionality of the HealthRisk AI Dashboard.

The video covers:

- Project Introduction
- Dashboard Navigation
- User Input Workflow

- Prediction Generation
- Risk Assessment Results
- HealthRisk Lab Simulation
- Deployment Demonstration

Video Link:

https://drive.google.com/file/d/138ewReRkYDNMr6FQB4DwGJ2ORap6KZ-o/view?usp=sharing

13.3 GitHub Repository

The complete source code, datasets, trained machine learning models, and project documentation are available in the GitHub repository.

Repository Link:

https://github.com/Madhuri-Kumari2504/HealthRiskAI

Chapter 14

Future Scope

- Real-Time Data Streams
- Mobile Application
- Explainable AI

Chapter 15

Conclusion

HealthRisk AI Dashboard demonstrates the practical application of Machine Learning in healthcare analytics.

The system successfully integrates multiple predictive modules into a single interactive platform and provides decision support capabilities for healthcare-related risk assessment.

Bibliography

[1] Scikit-Learn Documentation: https://scikit-learn.org

[2] Streamlit Documentation: https://streamlit.io

[3] Python Documentation: https://www.python.org

[4] Machine Learning for Healthcare Research Papers.